Written exam 20 December, 2017.

Course title: Network Optimization **Course no:** 42115

Aids allowed: Written aids permitted (textbooks, notes, old assignments etc.)

Exam duration: 4 hours

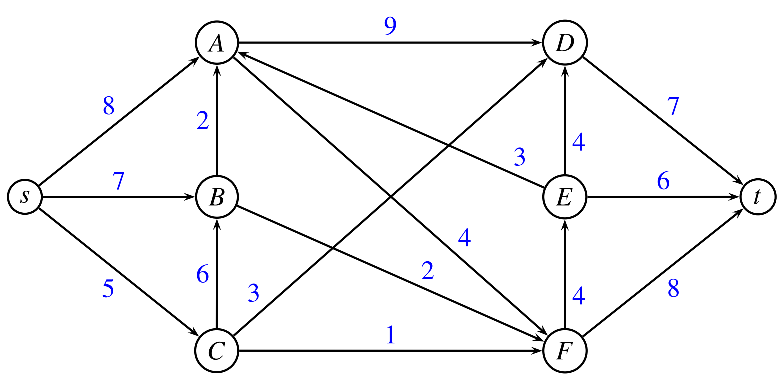
Weighting: Small problems give 2 points, medium problems give 3 points, large problems 4 points. A total of 42 points can be obtained.

Language: You can write in Danish or English.

Figures: All figures can be found on the last pages of this assignment. You can use them in your answer by drawing on the figure.

Maximum flow

Consider the following maximum flow problem where edge weights denote the capacity of an edge.



Problem 1: (medium) Solve the above maximum flow problem using an augmenting path algorithm. In each iteration report which augmenting path was used and the capacity. (You do not need to draw the residual graph in each iteration) ■

Problem 2: (medium) Draw the residual graph when the algorithm terminates. Indicate a minimum cut, and report the capacity of the cut and the value of the flow. ■

Transportation problem

Consider the following transportation problem. We have three suppliers A_1, A_2, A_3 and two customers B_1, B_2. The cost of transporting one unit from A_i to B_j is given in the following table

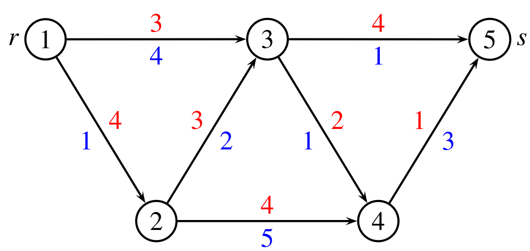
	B_1	B_2	supply
A_1	2	4	9
A_2	1	5	2
A_3	6	2	4
demand	7	8	

The supply from the A_i nodes is stated in the right column, while the demand in the B_j nodes is stated in the last row.

Problem 3: (large) Starting from a greedy solution, solve the problem using the network simplex algorithm. In each iteration report the dual variables and reduced costs. ■

Resource constrained shortest path problem

Consider the following directed acyclic graph. Every edge (i, j) has an associated cost c_{ij} (red) and time t_{ij} (blue). We are searching for a shortest path (with respect to cost) from r to s. There is a time constraint T imposed on the path. We would like to find the shortest path for every possible value of T.

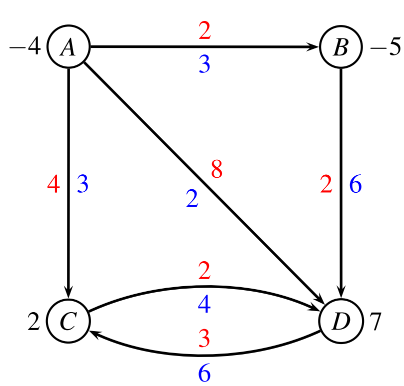


Problem 4: (small) Describe the algorithm you will use to solve the problem. ■

Problem 5: (large) Solve the problem, and report a table with shortest path from r to s for each possible value of T. ■

Min cost flow

Consider the following network with edge costs c_{ij} (red) and edge capacities u_{ij} (blue). The demand b_i (black) in every node is positive for a sink, and negative for a source.



Problem 6: (small) Find an initial solution by solving a maximum flow problem. ■

Problem 7: (large) Use the augmenting cycle algorithm to solve the min cost flow problem to optimality. In each iteration report the augmenting cycle, the cost of the cycle, and the new flow. In the last iteration explain the stop criteria. Also report the cost of the optimal solution. ■

Critical path problem

You have bought a new bicycle on the internet. However, it arrives in several pieces, with detailed instructions for assembling. Fortunately, you have many friends who are willing to help you, so several tasks can be carried out in parallel. The following table shows in which order the parts should be assembled, and how long time it takes:

Activity	Description	Predecessors	Duration
A	Chain	B,C	2
B	Crankset	E,H	7
C	Chainring	D	1
D	Handlebar	B,F,H	2
E	Spindle		2
F	Wheel	H,I	5
G	Pedals	C,J	4
H	Fork		3
I	Rear gear		3
J	Brakes	F	4

Problem 8: (small) Order the tasks in topological order. ■

Problem 9: (large) For each activity, find the earliest starting time and the latest starting time. Also, report the makespan. ■

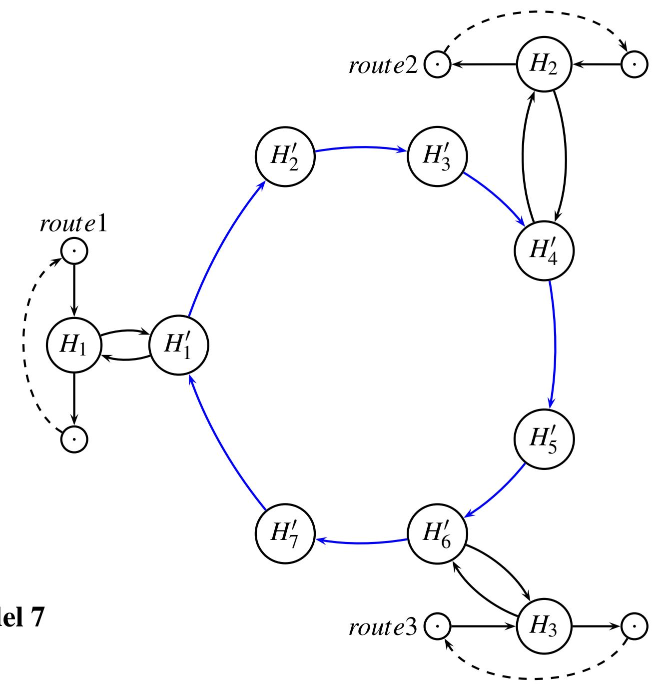
Problem 10: (small) Mark all critical paths. How many paths are there? ■

Arranging a dinner

Several families go out to dinner together. To increase their social interaction, they would like to sit at tables so that no two members of the same family are at the same table. Assume that the dinner contingent has p families, and that the i'th family has a_i members. Also assume that q tables are available and the j'th table has a seating capacity of b_j.

Problem 11: (large) Show how to formulate the problem as a maximum flow problem, and describe how the solution can be used to answer whether a seating is possible. You can use a small figure to illustrate your idea. ■

Modeling a hub



Model 7

We model the hub port H by using an extra node H_1, H_2 and H_3 for each route. Moreover, we introduce additional nodes $H'_1, H'_2, \ldots, H'_7$ representing the seven week days (1=Monday, 2=Tuesday, …, 7=Sunday). The latter nodes are connected in a cycle.

The crane capacity for loading/unloading is 10 containers for routes 1 and 2, while the capacity for loading/unloading is 7 containers for route 3 since it is operated by a smaller crane. There are separate cranes for loading and unloading, having the same capacity of 10 or 7.

The cost of a transshipment is 5$ for route 1 and 2, while it is 3$ for route 3. This mean that if you transship from route 1 to route 3 you pay 5$ for unloading the container at H and 3$ for loading the container.

Finally, the port has a limited storage capacity of $\overline{c} = 12$ containers on shore at any time. This means that the number of containers waiting one or more days for transshipment cannot exceed $\overline{c}$. The shipping company must pay 1$ for each container each day it is stored in the port.

It is assumed that, `route1` arrives Monday, `route2` arrives Thursday, and `route3` arrives Saturday.

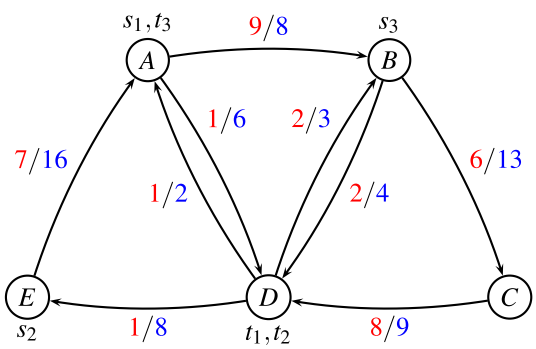
Consider Model 7 from Project Assignment 2017. The graph is used to model a hub port, where three routes meet and transship containers between the vessels.

It is now assumed that *all containers need to go through custom declaration immediately after unloading*. It takes two days to get a container through the custom declaration, so if the vessel arrives at day t the container will be available day $t+2$. The custom has infinite capacity for storing cotainers in a separate stockpile, and it costs 1\$ per day in storage cost. Containers that stay on board a vessel do not need to go through custom declaration.

Problem 12: (large) Modify the above graph to reflect the times for loading/unloading. Is it possible to model transshipment costs, transshipment time, crane capacities, storage capacities, and storage costs? For those parameters where you answer yes, indicate the values on the graph. For those parameters where you answer no, argue why not. ■

Flowing containers in a network

MEDITLINE **(Route net of a Mediterranean liner shipping company)**



k	s_k	t_k	d_k	w_k
1	A	D	3	1
2	E	D	4	2
3	B	A	2	4

For edge (i, j) the first number (red) is the cost c_{ij} and the second number (blue) is the capacity u_{ij}. Cities are $A =$ Alexandria, $B =$ Barcelona, $C =$ Casablanca, $D =$ Dubrovnik, $E =$ Ephesus.

Consider the above weighted multi-commodity flow problem. All demands and costs are exactly as in the Project Assignment 2017, hence we have the possible routes r outlined in the below table. The cost of sending 1 unit of size w_k along a route r is calculated as $w_k \sum_{(i,j)\in r} c_{ij}$.

number	commodity k	route r	w_k	$\sum c_{ij}$	cost
1	1	$ABCD$	1	23	23
2	1	ABD	1	11	11
3	1	AD	1	1	1
4	2	$EABCD$	2	30	60
5	2	$EABD$	2	18	36
6	2	EAD	2	8	16
7	3	$BCDEA$	4	22	88
8	3	$BDEA$	4	10	40

We solve the problem through column generation. Instead of starting from some dummy routes, we guess a part of the solution, by using route $r = 3$ for commodity 1, and routes $r = 7, 8$ for commodity 3. Hence, we only need to have a dummy route for commodity 2. This dummy route is denoted route 0, and it goes directly from E to A without using any capacity.

This leads to the following LP-model, the (*restricted master problem*):

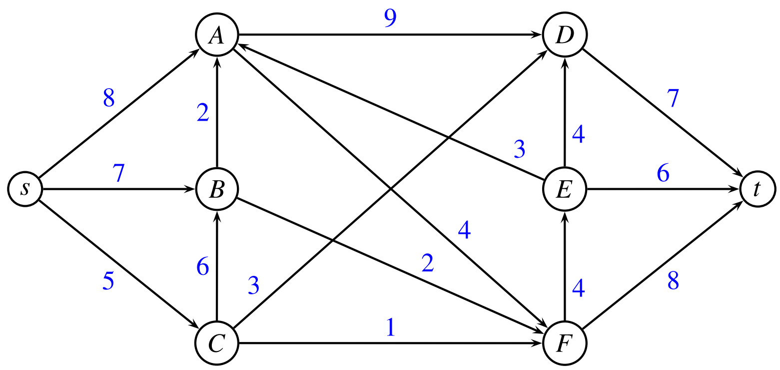
$$
\begin{array}{llllllll}
\min & 1x_3 & + \ 88x_7 & + \ 40x_8 & + \ 100x_0 & & & \\
\text{s.t.} & & & & & \leq 8 & (y_{AB}) & \\
 & x_3 & & & & \leq 6 & (y_{AD}) & \\
 & & x_7 & & & \leq 13 & (y_{BC}) & \\
 & & & x_8 & & \leq 4 & (y_{BD}) & \\
 & & x_7 & & & \leq 9 & (y_{CD}) & \\
 & & & & & \leq 2 & (y_{DA}) & \\
 & & & & & \leq 3 & (y_{DB}) & (1) \\
 & & x_7 \ + & x_8 & & \leq 8 & (y_{DE}) & \\
 & & x_7 \ + & x_8 & & \leq 16 & (y_{EA}) & \\
 & x_3 & & & & \geq 3 & (\overline{y}_1) & \\
 & & & & x_0 & \geq 4 & (\overline{y}_2) & \\
 & & x_7 \ + & x_8 & & \geq 2 & (\overline{y}_3) & \\
\multicolumn{8}{l}{x_3, x_7, x_8, x_0 \geq 0}
\end{array}
$$

An optimal solution to the above problem is $x_3 = 3$, $x_7 = 1$, $x_8 = 1$, $x_0 = 4$ with objective $z = 531$. The dual variables are $y_{BD} = -12$, $\overline{y}_1 = 1$, $\overline{y}_2 = 100$, $\overline{y}_3 = 88$. All other dual variables are zero.

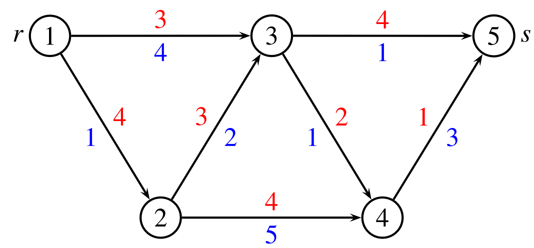
Problem 13: (large) Calculate the reduced costs of all routes not in the restricted master problem, and indicate which route should be added to the problem in the next iteration. ■

end of assignment

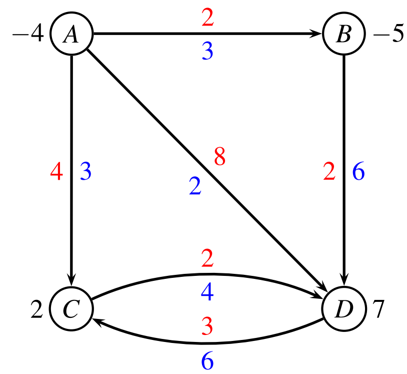
Maximum flow



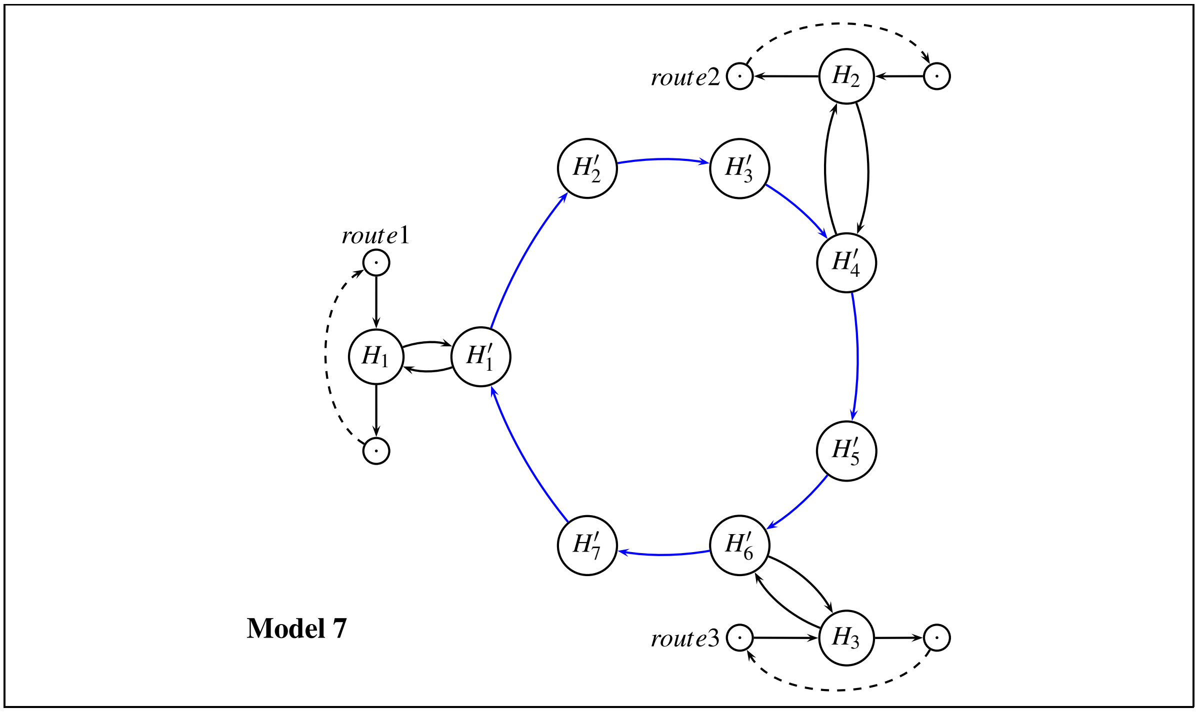
Resource constrained shortest path problem



Min cost flow



Modeling a hub



Model 7